STAT6 | BUSINESS MANAGEMENT SYSTEM (BMS)

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Comprehensive Product Feature, Technical Architecture & Implementation Specification

Complete formal reference manual for the unified sovereign business operating platform, 14 autonomous microservices, 4D-RBAC security model, and web front-end engine.

PRODUCT NAME	STAT6 BMS Enterprise	**OFFICIAL TAGLINE**	"We Shall Collaborate"
ARCHITECTURE MODEL	Autonomous Microservices Mesh (:8000–:8013)	**PRIMARY DATABASE**	Supabase Cloud PostgreSQL 17.6 (52 Tables)
SECURITY & AUTH	Native OAuth 2.0 (Password Grant) + RS256 JWT	**CRYPTOGRAPHIC VAULT**	PBKDF2-HMAC-SHA256 (600k iter) + AES-256-GCM
FRONTEND STACK	React 18.3, TypeScript 5.7, Tailwind CSS, Vite 6.4	**VERIFICATION STATUS**	67/67 Automated Assertions Passed (100%)

1. EXECUTIVE PRODUCT SUMMARY & CORE ARCHITECTURE

1.1 Mission & Product Thesis

STAT6 BMS is an enterprise-grade Business Management System engineered to eliminate the severe fragmentation, vendor lock-in, and synchronization lag inherent in legacy SaaS ecosystems (specifically Zoho One, HubSpot, archaic Deluge scripts, and disparate point solutions). Under the guiding architectural mandate *"We Shall Collaborate"*, STAT6 provides a unified, sovereign, zero-3rd-party-leakage operating platform connecting commercial CRM, agile sprint execution, double-entry general ledgers, multi-warehouse inventory, procurement, and HR payroll into a mathematically reconciled single source of truth.

1.2 The Zoho One / Legacy Enterprise Disconnect

EVALUATION DIMENSION	LEGACY FRAGMENTED SAAS (E.G., ZOHO ONE / HUBSPOT)	STAT6 BMS SOVEREIGN ARCHITECTURE
Data Synchronization	Delayed polling sync (5–15 min intervals); broken webhooks between Zoho CRM, Books, and Creator; phantom duplicates.	**Zero-Sync Canonical Schema:** Single PostgreSQL 17 database with ACID transactions and immediate internal event-bus dispatch.
Automation & Logic	Archaic proprietary scripting (Deluge Scripting) with strict execution governor limits, untyped syntax, and fragile debugging.	**Visual DAG Workflows & Python Services:** Strongly typed Python/FastAPI microservices, topological graph execution, and natural-language AI generation.
Financial Integrity	Opaque journal entries without cryptographic hash verification; external exports susceptible to tampering.	**Immutable Double-Entry Ledger:** Cryptographic SHA-256 hash chaining on all journal entries; enforced mathematical balance constraint (Σ Debit = Σ Credit).
User Access & Cost	Per-user, per-portal license penalization; external vendors and clients require expensive seat upgrades.	**7 Free Portal Personas:** Unlimited free customer, vendor, and partner portals guarded by token-based tenant isolation.
Security & Data Privacy	3rd-party telemetry, cross-site trackers, external Auth0/Firebase dependencies leaking enterprise tenant metadata.	**Native OAuth 2.0 & Sovereign Vault:** Self-hosted OAuth 2.0 gate, PBKDF2-HMAC-SHA256 master secrets, AES-256-GCM field encryption, zero tracking.

1.3 Microservices Topology & Internal Bus

STAT6 is structured as an interconnected mesh of 14 autonomous services orchestrated by an intelligent API Gateway (Port 8000). Every service operates independently with isolated domain boundaries, schema ownership, and event publishers:

- **API Gateway (`:8000`)**: Reverse proxy, SSL termination, JWT bearer token verification, RBAC context injection, and health telemetry.
- **Autonomous Microservices (`:8001` – `:8013`)**: Covering Auth, CRM, Projects, Marketing, HR, Finance, Inventory, Procurement, Support, Collaboration, Compliance, Workflows, and AI Copilot.
- **Event-Driven Choreography**: Powered by an asynchronous `WHEN-IF-DO` event bus. When a business event occurs (e.g., `DEAL_WON`), downstream actions (generating a draft invoice, allocating inventory, triggering a vendor RFQ, and alerting staff) execute with zero human intervention.

2. FRONT-END ENGINEERING & LANDING HERO ARCHITECTURE

2.1 Procedural Atmospheric Bowl-Glow Canvas Engine

The public entrance to STAT6 features a custom procedural atmosphere rendered via vanilla HTML5 Canvas 2D without 3rd-party graphical runtimes. To ensure 60 FPS rendering on any device, computation is executed on an ultra-light **96×60 pixel canvas** and bilinearly scaled across the entire viewport:

```
// Mathematical Formulation of Atmospheric Glow Drift:
const driftTip  = curTip  + Math.sin(time * 0.00033 + 1.3) * 0.02;
const driftHalf = curHalf + Math.sin(time * 0.00050) * 0.015;
// Bilinear upscaling with CSS 'inset: 0; width: 100%; height: 100%; pointer-events: none;'
// Generates a soft, breathing sky-cyan (#8fd5f1) bowl hanging seamlessly from the top edge.
```

2.2 Pinned Typography, Responsive Wrapping & Letter-by-Letter Reveal

The landing experience transitions through three distinct scroll phases, engineered to eliminate layout shifts, horizontal clipping, and sticky failure:

Scene 1 & 2: Hero & Pinned Statement

Word "system" & Headline Visibility: Configured with `clamp(28px, 3.8vw, 56px)` and responsive boundaries (`max-w-[min(880px, 86vw)]`) to prevent overflow.
Sticky Pinning via `overflow-x: clip` : Avoids `overflow-x: hidden` which breaks CSS sticky contexts. The statement stays centered (`top: 0; height: 100vh`) across scroll points 600px to 2200px.
Letter Painter Engine: Dynamically splits characters into DOM spans once (guarded against React Strict Mode re-mounts) and maps scroll progress: `pStmt = clamp((q - 0.32) / 1.15)` .

Scene 3: Sovereign BMS Console

Removal of Irrelevant Media: The stock photo of a container ship was completely replaced with the authentic Sovereign BMS Operating Console.
Real-Time Live Telemetry: Features 3 live KPI cards (Paid Revenue MTD, Reconciled GL, RBAC Guard with 7 portals), a 6-node architecture matrix, an audit event ledger stream, and 1-click test persona logins.
Hover Interaction: Large pills feature glowing cyan dots (`#2fe0d0`) that expand into a 32px circular chevron on mouse hover.

2.3 Dual-Mode High-Contrast Theme Engine

The platform includes a zero-flicker dark/light theme engine configured for strict contrast compliance:

- **Dedicated Icon Toggle:** Replaced raw text links with Lucide `Sun` (amber) and `Moon` (blue) icons located in the top header, the sidebar footer, and the floating pill navbar.
- **State Synchronization & Persistence:** Synchronizes with `localStorage.getItem('stat6_theme')` and toggles `document.documentElement.classList.toggle('dark')` immediately on hydration.
- **Global Heading Safeguards:** Overhauled CSS specificity so that `.dark h1, .dark h2, .dark h3, .dark h4, .dark h5, .dark h6` are strictly forced to `#ffffff !important` , eliminating black-on-black text bugs.
- **Button Surface Safeguards:** Enforces `.dark button.bg-white { background-color: #1a2744 !important; color: #cbd5e1 !important; }` to eliminate blank white pill buttons in dark mode.

3. 4D-RBAC SECURITY ARCHITECTURE & THE 7 PERSONA PORTALS

3.1 The 4D-RBAC Policy Engine

Traditional RBAC models operate on flat user-to-role mappings. STAT6 introduces a **4-Dimensional Evaluation Tuple** evaluated on every inbound API call:

```
f(Access) = Policy(Role × Persona × Module × Action) ∈ { ALLOW, DENY, MASK_DATA, READ_ONLY }
```

This guarantees that external users (such as Vendors or Customers) are cryptographically restricted to their own tenant records without having visibility into internal financials, HR payroll, or other vendor bids.

3.2 Technical Specification of the 7 Supported Personas

PERSONA IDENTIFIER	TARGET AUDIENCE	ACCESSIBLE MODULES & PERMISSIONS	DATA ISOLATION & SECURITY CONTROLS
01. ADMIN `Internal Full`	Enterprise Super Admins, CXOs, IT Directors	Unrestricted access across all 14 modules. Dynamic module toggle switches, 4D-RBAC policy editor, database backup & audit trails.	MFA required; PBKDF2 vault master passkey; immutable activity logging.
02. MANAGER `Internal Head`	Department Heads, VPs of Sales/Ops, Controllers	CRM Deals approvals, Project milestone sign-offs, PO expenditure approvals up to $100k, team KPI metrics, SLA escalation alerts.	Departmental scope; cannot toggle system modules or edit raw SQL tables.
03. EMPLOYEE `Internal Self`	Staff Members, Engineers, Account Reps	Assigned Jira tasks, GPS geofenced attendance logs, personal statutory payslip downloads, expense receipts upload, team chat.	Strictly limited to own timesheets, salary records, and assigned sprint tickets.
04. CUSTOMER `External Portal`	Enterprise Clients, Buyers, Procurement Leads	Review CPQ proposals, 1-click legal E-sign, download GST invoices, settle payments via Stripe/ACH, submit support tickets.	Multi-tenant row-level security (RLS); isolated to quotes/invoices where `client_id = token.sub`.
05. VENDOR `External Portal`	Suppliers, Contractors, Logistics Providers	View allocated Purchase Orders (PO), acknowledge delivery schedules, upload delivery slips, submit RFQ bids for open tenders.	Blind bidding isolation: cannot view competitor RFQ bids or corporate balance sheets.
06. PARTNER `External Portal`	Channel Partners, Resellers, Co-Sellers	Register new partner leads, lock 15% partner commission margins, access marketing collateral, review payout statements.	Deal registration conflict detector; automatic 30-day account protection lock.
07. AUDITOR `Compliance Vault`	SOC 2 Auditors, ISO 27001 Inspectors, Tax CPAs	Time-boxed read-only compliance vault, SOC 2 control matrices, cryptographic ledger hash verification, GDPR DSAR logs.	Time-decaying access tokens; zero write/mutation permissions; watermarked PDF exports.

4. TECHNICAL SPECIFICATION OF ALL 14 ENTERPRISE MODULES

4.1 Module 01: Commercial CRM, Deals Pipeline & CPQ E-Sign (Port 8002)

- **Leads & Opportunity Pipeline:** Kanban drag-and-drop deal management with stages (*Lead In, Discovery, Proposal Sent, Negotiation, Closed Won, Closed Lost*). Deals over $50,000 trigger automated risk mitigation reviews.
- **Configure-Price-Quote (CPQ) Engine:** Real-time itemized quoting with unit rates, volume discounts, dynamic margin calculations, and PDF generation.
- **Native Cryptographic E-Sign:** Implements legal e-signatures backed by SHA-256 document hashing, signer IP recording, and audit trails without DocuSign fees.
- **Appointment Scheduling Links:** Slot-based appointment booking system with automated timezone normalization and Google Calendar/Outlook ICS generation.

4.2 Module 02: Operations & Jira Agile Sprints (Port 8003)

- **Agile Sprint Lifecycle:** Full sprint planning, active sprint tracking, and sprint retrospective closure with automatic rollover of incomplete user stories.
- **Velocity Burndown Algorithm:** Computes daily story-point burn rate against ideal linear velocity; detects team capacity bottlenecks in real time.
- **Critical Path Method (CPM) & Gantt Dependencies:** Algorithmic forward/backward pass solver identifying critical delivery milestones and slack time.
- **Telemetry & PR Integration:** Native linkage of tickets (e.g., `FIN-52`) to GitHub Pull Requests, Figma designs, and Honeycomb tracing.

4.3 Module 03: Marketing Automation & Multi-Touch Attribution (Port 8004)

- **Multi-Touch Revenue Attribution Engine:** Mathematically attributes closed-won CRM revenue across marketing channels using 5 distinct models:

```
1. First-Touch (100% Acquisition)       4. U-Shaped (40% First, 40% Lead Creation, 20% Middle)
2. Last-Touch (100% Conversion)        5. Time-Decay (Exponential 7-day half-life weighting)
3. Linear (Equal multi-touch division)
```

- **Drip Journeys:** Trigger-based email automation sequences (e.g., *Enterprise Trial Onboarding*) with condition branches and delay timers.
- **UTM Campaign Broadcaster:** Automated generation of tracked short-links and broadcast deliverability analytics (Email, SMS, WhatsApp).

4.4 Module 04: HR & Staff Directory, Geofenced Attendance & Statutory Payroll (Port 8005)

- **GPS Geofenced & IP-Restricted Attendance:** Validates mobile check-ins using the Haversine spherical distance formula: `d = 2r · arcsin(√(sin²(Δφ/2) + cos(φ1)cos(φ2)sin²(Δλ/2)))`. Rejects punches outside authorized corporate boundaries.
- **Indian Statutory Payroll Calculator:** Automated calculation of Provident Fund (PF: 12% employee + 12% employer), Employee State Insurance (ESI: 0.75% / 3.25%), Professional Tax (PT slab), and TDS withholding with formal payslip PDF compilation.
- **OKRs & Applicant Tracking System (ATS):** Cascading departmental objectives and full candidate hiring pipeline from resume parsing to offer generation.

4.5 Module 05: Finance & Invoices, Double-Entry General Ledger & AI Expense OCR (Port 8006)

- **Mathematical Double-Entry Balance Constraint:** Every transaction enforces strict ledger equilibrium (Σ Debits = Σ Credits). Transactions failing equilibrium are rejected with HTTP 422.
- **Cryptographic SHA-256 Audit Chain:** Every journal row includes `prev_hash` and `curr_hash = SHA256(id + date + accounts + amount + prev_hash)`, making historical ledger tampering impossible.
- **AI Receipt Scanner & Policy Enforcement:** Multi-modal OCR extracting merchant name, tax ID, line items, and currency, cross-checking corporate expense limits.
- **GST / VAT Tax Compliance Engine:** Automated split computation for intra-state (CGST + SGST) and inter-state (IGST) invoices with e-invoicing JSON generation.

4.6 Module 06: Multi-Warehouse Inventory, Batches, Serials & FIFO Valuation (Port 8007)

- **Multi-Location Warehouse Management:** Granular inventory tracking across physical warehouses, aisle zones, and bin coordinates.
- **Lot, Batch & Serial Tracking:** Enforces First-In, First-Out (FIFO) and Weighted Average Cost (WAC) models; full recall traceability for perishable goods.
- **Fixed Asset Register & Depreciation:** Automated monthly straight-line and declining-balance depreciation journal entries posted straight to Finance.

4.7 Module 07: Procurement, Vendor Management & 3-Way Match (Port 8008)

- **Automated 3-Way Invoice Matching:** Cryptographically reconciles *Purchase Order (PO)* ↔ *Goods Receipt Note (GRN)* ↔ *Vendor Invoice*. Discrepancies exceeding a 1.5% variance tolerance lock disbursement automatically.
- **Vendor RFQ Tender Matrix:** Side-by-side comparison matrix evaluating competing supplier bids by price, lead time, warranty terms, and historical SLA performance.

4.8 Module 08: Customer Support Desk, 360° Account Context & Knowledge Base (Port 8009)

- **Ticket Queues & Dynamic SLA Timers:** Tiered SLA response clocks (Urgent: 1 hr, High: 4 hrs, Medium: 24 hrs) with automated escalation triggers.
- **360° Customer Context Sidebar:** Displays live CRM deal value, active delivery projects, outstanding invoices, and contract terms directly adjacent to the ticket thread.
- **Knowledge Base CMS:** Searchable repository of public and internal troubleshooting guides, reducing tier-1 support ticket volume.

4.9 Module 09: Executive BI Telemetry, Predictive Forecasting & Report Builder (Port 8010)

- **Cross-Module 360 Matrix:** Zero-sync canonical SQL join querying Sales, Delivery, Finance, and Support in a single high-performance query.
- **Predictive 3-Month Cash Flow Forecast:** Algorithmic projection based on historical receivables velocity, sales pipeline stage probability, and committed supplier payables.
- **Low-Code Custom Report Builder:** Drag-and-drop report creator allowing non-technical managers to construct cross-departmental tabular reports.

4.10 Module 10: Team Collaboration, Internal Channels & Announcements (Port 8011)

- **Internal Chat Channels:** Real-time team messaging organized by department and delivery project, replacing Slack/Teams sprawl.
- **Company-Wide Broadcast Announcements:** Pinned executive memos with mandatory read-receipt tracking.
- **Shared Enterprise Calendar & Notes:** Consolidated view of sprint deadlines, milestone releases, payroll dates, and private scratch notes.

4.11 Module 11: Compliance, SOC 2 Type II, Secret Vault & GDPR DSAR (Port 8012)

- **12 SOC 2 Type II Continuous Controls:** Real-time compliance monitoring covering TLS 1.3 enforcement, least-privilege RBAC, audit log retention, and daily backups.
- **PBKDF2-HMAC-SHA256 Encrypted Password Vault:** Enterprise credential manager with master-key derivation (600,000 iterations) and AES-256-GCM encrypted secret payloads.
- **GDPR DSAR Export & Right-to-be-Forgotten:** 1-click JSON export of all personal identifiable information (PII) and irreversible cryptographic data anonymization routines.

4.12 Module 12: Visual DAG Workflow Builder & 1-Click Zoho One Importer (Port 8013)

- **Natural-Language AI Workflow Generator:** Compiles natural language prompts (e.g., *"When deal > $50k won, require finance approval then invoice"*) into executable Directed Acyclic Graphs (DAGs) without writing a line of code.
- **Unified Approvals Engine:** Centralized multi-tier approval inbox across HR leave, PO expenditures, invoice write-offs, and custom workflows.
- **1-Click Zoho One Migration Importer:** Automated ingestion engine parsing exported Zoho CRM, Books, Creator, and Desk CSV/JSON archives into canonical STAT6 relational schemas.

4.13 Module 13: STAT6 AI Copilot & Autonomous Agentic Framework (Port 8013)

- **Proactive 8:00 AM Morning Executive Briefing:** Synthesizes yesterday's closed deals, daily revenue vs quota, critical support escalations, and overdue sprint milestones into a 2-minute actionable summary.
- **RFM Customer Churn & Anomaly Detector:** Evaluates Recency, Frequency, and Monetary scores to identify enterprise accounts showing disengagement signals before they churn.
- **1-Click Contract, RFP & Ticket Thread Summarizer:** Powered by Google Gemini models; extracts liability clauses, indemnity caps, payment terms, and support root-causes in seconds.

4.14 Module 14: Organization Settings & Low-Code Custom Object Builder

- **Dynamic JSON Schema Custom Field Engine:** Allows administrators to attach custom typed fields (strings, dates, formulas, dropdowns) to any native entity without database migrations.
- **Low-Code Custom Object Creator:** Enables creation of custom business objects (e.g., *Equipment Fleet*, *Vessel Log*) with auto-generated relational tables and CRUD views.
- **14 Modular Feature Toggles:** Granular master switches allowing organizations to selectively enable or disable individual modules based on their deployment footprint.

5. SECURITY, CRYPTOGRAPHY & DATA GOVERNANCE

5.1 Authentication, Tokens & Identity Management

STAT6 enforces a sovereign security perimeter completely independent of 3rd-party auth brokers:

- **OAuth 2.0 Password Grant Specification:** Compliant with RFC 6749; exchanges validated enterprise credentials for cryptographically signed JSON Web Tokens (JWT).
- **Cryptographic Signature:** Signed using RS256/HS256 with 24-hour expiration windows and automated refresh token rotation.
- **Password Hashing:** Passwords hashed using standard PBKDF2-HMAC-SHA256 with 600,000 rounds and unique 32-byte cryptographic salts.

5.2 Data Encryption at Rest & in Transit

LAYER / ASSET	CRYPTOGRAPHIC IMPLEMENTATION	STANDARD & COMPLIANCE TARGET
Data in Transit	TLS 1.3 mandatory; HSTS with 31536000s max-age; perfect forward secrecy.	FIPS 140-3, PCI-DSS v4.0 Requirement 4.1
Secret Vault at Rest	Envelope encryption: Master Key Derived via PBKDF2; payloads encrypted with AES-256-GCM.	NIST SP 800-132, SOC 2 Type II Confidentiality
Financial Ledger Chain	Immutable SHA-256 hash chaining: `curr_hash = SHA256(row + prev_hash)`.	SOX Section 404, AICPA Trust Services Criteria
E-Signature Evidence	SHA-256 document manifest combined with signer IP, user-agent, timestamp, and signature glyph.	ESIGN Act (15 U.S.C. § 7001), eIDAS Regulation

5.3 Database Governance & Supabase Integration

The production persistence tier utilizes **Supabase Cloud PostgreSQL 17.6** hosted in the EU (Frankfurt / `eu-central-1`):

- **52 Normalized Relational Tables:** Enforcing foreign key constraints, check constraints, and unique compound indexes across all business entities.
- **Multi-Tenant Isolation:** Guarded by PostgreSQL Row-Level Security (RLS) policies ensuring tenant partition isolation.
- **Automated Snapshot Backups:** Point-in-time recovery (PITR) with continuous write-ahead logging (WAL) and automated off-site archival.

6. CORE API ENDPOINT DIRECTORY & VERIFICATION METRICS

METHOD	ENDPOINT URI	TARGET MICROSERVICE	FUNCTIONALITY & OUTPUT
`POST`	`/auth/oauth/token`	Auth Service (:8001)	RFC 6749 OAuth 2.0 password grant; issues JWT access token.
`GET`	`/crm/deals`	CRM Service (:8002)	Retrieves categorized deals pipeline with stage probabilities.
`POST`	`/crm/quotes`	CRM Service (:8002)	Generates itemized CPQ quotation with cryptographic e-sign link.
`GET`	`/projects/sprints/active`	Projects Service (:8003)	Active Jira sprint board, velocity burndown, and task cards.
`GET`	`/marketing/attribution`	Marketing Service (:8004)	Computes multi-touch revenue attribution across ad spend channels.
`POST`	`/hr/attendance/punch`	HR Service (:8005)	GPS geofenced and IP-restricted mobile attendance check-in.

METHOD	ENDPOINT URI	TARGET MICROSERVICE	FUNCTIONALITY & OUTPUT
`POST`	`/hr/payroll/calculate`	HR Service (:8005)	Calculates statutory PF, ESI, Professional Tax, TDS & net salary.
`POST`	`/finance/ledger/entry`	Finance Service (:8006)	Posts double-entry balanced transaction with SHA-256 hash chaining.
`POST`	`/procurement/3way-match`	Procurement Service (:8008)	Reconciles PO vs Delivery Slip vs Invoice; checks variance tolerance.
`GET`	`/support/tickets/360`	Support Service (:8009)	Retrieves customer ticket queue with attached CRM & finance context.
`GET`	`/bi/predictive-forecast`	BI Service (:8010)	3-month cash flow and revenue trajectory with confidence bands.
`POST`	`/compliance/vault/store`	Compliance Service (:8012)	Stores AES-256-GCM encrypted credential into sovereign secret vault.
`POST`	`/automation/dag/compile`	Automation Service (:8013)	Compiles natural-language prompt into executable DAG workflow.
`GET`	`/copilot/morning-briefing`	AI Copilot (:8013)	Generates proactive executive morning briefing from live DB state.

PRODUCTION VERIFICATION & QUALITY ASSURANCE SUMMARY

All 14 microservices and front-end user interfaces have completed rigorous automated verification:

- **Test Execution:** 11 comprehensive automated test suites executed via `run_all_tests.py`.
- **Assertion Rate: 67 / 67 Assertions Passed (100.0% Pass Rate)** across OAuth, Double-Entry GL, 3-Way Match, GPS Geofencing, and RBAC.
- **Frontend Build:** Vite 6.4 + TypeScript 5.7 compiled in **4.26s with 0 errors**.
- **Deployment Targets:** Configured for instant deployment to Vercel (Front-end) and Google Cloud / Supabase (Back-end).

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